

Music-Gen — M-EAR-1/feature-representation-audit (cycle 1, fork dc8cba4b79eb, clone 1)

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Abstract

Cycle 1 of clone 1 discharged the last cheap Path A probe on the ear-model chassis. Cycle 22 clone-2 falsified the cycle-6 CORN 1-7 head chassis on $N = 55$ synthetic labels at $\tau \geq 0.7$; cycle 23 clone-1 falsified three orthogonal regularized head variants at the relaxed $\tau \geq 0.4$ bar. This branch tested the remaining head-side hypothesis’s mirror — whether the cycle-6 head architecture over a slimmer feature representation would produce recipe-invariant rank predictions under the same frozen instrument. Two representations were evaluated (HEUR-only 4-D and PANNs-only 2048-D) under the UNCHANGED cycle-22 stability-audit harness with SHA-anchored invariance and the UNCHANGED cycle-6 CORN head architecture (only D_{in} changing at instantiation); VGGish-only R3 was legitimately deferred because `has_vggish=False` on the cache and running the extractor over 55 clips is out-of-scope per the brief. Both representations **FAIL C2’**: HEUR-only mean $\tau = -0.076$ (bimodal span $[-0.958, +0.951]$); PANNs-only mean $\tau = +0.006$. C3’ byte-determinism $\times 2$ PASS uniformly (heur ec429bdf...5e8c, panns f98a498c...d39e). HEUR-only C1’ PASS (best MAE 0.782 beats the cycle-6 anchor 0.891) is scientifically interesting but not a Path A rescue — it is the underdetermined-regressor signature at extreme low D , not evidence for a HEUR-first real-label recipe. The pre-registered “no representation PASSES C2’” interpretation rule fires cleanly; the sub-milestone closes at **invalidated/high** and the recommendation to commit Path B (defer all ear calibration to post-egress real labels) at cycle 26 is the pre-registered outcome. Path A on the ear-model chassis at $N = 55$ synthetic labels is now closed comprehensively across three orthogonal design axes.

Introduction

By the end of cycle 23 two consecutive VALIDATED audits under the same frozen SHA-anchored / byte-determinism $\times 2$ methodology had invalidated their pre-registered hypothe-

ses on $N = 55$ synthetic labels. Cycle 22 established that no cycle-6-shape head can pass at $\tau \geq 0.7$; cycle 23 established that no head chassis regularization (weight decay + dropout; bottleneck width; feature-rank reduction) can pass at the relaxed $\tau \geq 0.4$ bar. Cycle 23's auditor lean was Path B (defer to post-egress real labels) on a corpus-size reading of the flat τ -vs-MAE frontier; the researcher's cycle-25 decision was to spend one final cheap Path A probe on the feature side before committing to Path B, because the head-side sweep and the feature-side sweep are orthogonal design axes and a positive result on the feature side would legitimately reopen the head-side-fix hypothesis under a different (more informative) feature distribution. This branch is that probe with the rubric locked pre-run and the interpretation rules pre-registered.

Approach

Two representations at two extreme dimensions of the frozen feature cache.

- **HEUR-only 4-D** — the M-HEUR-1 mess-scale vector alone. Slim end of the design space; tests whether the ordinal signal (if any) lives in the hand-designed heuristics rather than the deep-net embedding.
- **PANNs-only 2048-D** — the PANNs Cnn14 penultimate embedding alone. Wide end of the design space; tests whether the ordinal signal (if any) lives in the AudioSet-pretrained embedding rather than the heuristics.
- **VGGish-only 128-D** — R3 deferral, honest and rule-consistent: the cache has `has_vggish=False`, `vggish_embed.shape=(0,)` because cycle-6 clone-2 chose not to invoke `use_vggish=True`, and running the extractor over 55 clips is explicitly out-of-scope per the brief §2. The frontier plot carries a deferral-marker row so future cycles can revisit without re-litigation.

Instrument invariance. All six cycle-22 harness anchor SHAs verified equal at run start (`stability_audit.py`, `synthetic_labels.py`, `stability_metrics.py`, `model.py`, `corn.py`, `features.py`); harness file SHA verified at run start AND end. The cycle-6 CORN head architecture is imported unchanged — only `D_in` differs at instantiation (4 for HEUR-only, 2048 for PANNs-only). No PRNG (AST-checked); no `sidcar_nonfactor` imports (AST-checked); interpreter guard on every new module; single-thread BLAS pins throughout.

Feature cache invariance (with concurrent-clone interference disclosed and mitigated). The audit's read set is exactly the 55 valset clips; those clip files are byte-identical pre/post. Sibling clone-0 concurrently wrote `gen_first_gen_*.npz` files into `data/ear/features/` during the run, which changed the all-files SHA manifest but did not touch any of the 55 valset clips. The driver added a valset-scope filter and preserved both manifests (`feature_cache_pre_post_shas.json` + `prior_all_files_*` fields) for auditability. The operative invariance property — the 55 valset clips are byte-identical between pre and post — holds.

Rubric locked pre-run (matches cycle-23's relaxed thresholds).

- **C1'** — MAE reproducibility against the constant cycle-6 anchor 0.891 (matching cycle-22 clone-2's actual methodology; cycle-6 anchor is a distinct out-of-namespace PC1+noise construction, not salt = 0).
- **C2'** — mean Kendall τ across the 10 SHA-256-salted synthetic recipes ≥ 0.4 .
- **C3'** — byte-determinism $\times 2$ on `stability_report_v3_<rep>.json`.

Pre-registered interpretation rules.

1. Any representation PASSES C2' $\rightarrow$ cycle 26 refines that feature family.
2. No representation PASSES C2' $\rightarrow$ cycle 26 commits to Path B (defer all ear calibration to post-egress real labels) with the strongest possible negative-finding justification.

Findings

Per-representation verdicts (both FAIL C2')

Representation	C1' (MAE anchor)	C2' (mean $\tau \geq 0.4$)	C3' (byte-det $\times 2$)	Overall
HEUR-only 4-D	PASS (best MAE 0.782 beats 0.891 anchor)	FAIL (mean $\tau = -0.076$; span $[-0.958, +0.951]$)	PASS (ec429bdf...5e8c)	FAIL
PANNs-only 2048-D	FAIL	FAIL (mean $\tau = +0.006$)	PASS (f98a498c...d39e)	FAIL
VGGish-only 128-D	—	—	—	DEFERRED (R3)

Both R1 and R2 fail C2' by very wide margins. The pre-registered “no representation PASSES C2'” interpretation rule fires cleanly.

The one substantive risk — HEUR-only C1' PASS — is not a Path A rescue

The HEUR-only representation can fit individual synthetic-label recipes tightly (best MAE $0.782 < \text{cycle-6 anchor } 0.891$), so C1' passes on that axis. But its mean τ is -0.076 with a symmetric bimodal span of $[-0.958, +0.951]$. That shape has a specific interpretation: the head learns a near-perfect ordering on each recipe individually, but the orderings it learns are near-orthogonal across recipes — a positive-almost-1 τ on one recipe and a negative-almost-1 τ on another cancel to a mean near zero. This is the **underdetermined-regressor signature at extreme low D** — 4 features are enough to fit any synthetic ordering the recipes generate, and the fit picks a different direction each time — not evidence for a HEUR-first real-label recipe. The report's §7.1 flags this correctly and does not spin C1' PASS as a partial positive.

τ -vs-MAE frontier is exhaustively negative across three orthogonal design axes

frontier_summary.json has 7 rows: cycle-6 baseline + cycle-23 three head-regularization variants + cycle-25 two representations + one R3 deferral marker. All 6 tested design points cluster near the $\tau \approx 0$ axis. There is no design point on the frontier that clears $C2' = 0.4$; the head-regularization axis (cycle 23) failed at $\tau \approx 0.06$ – 0.08 , the feature-representation axis (this cycle) failed at $\tau = -0.08$ and $+0.01$, and the frontier's shape — a tight cluster near zero across two orthogonal design axes — is the strongest empirical signal the campaign can produce without real labels that the ordinal information simply is not in the $N = 55$ synthetic-label regime.

Anchor invariance held

- Harness anchor SHAs: 6/6 byte-identical to cycle-22 values, verified at run start AND run end.
- Feature cache (valset scope): 55 clips byte-identical pre/post. All-files manifest changed due to concurrent clone-0 writes; both manifests preserved for auditability.
- Cycle-6 CORN head architecture: imported unchanged; only `D_in` differs at instantiation.
- No PRNG (AST-checked, 5 forbidden tokens).

- No sidecar_nonfactor imports (AST-checked).
- Interpreter guard `assert sys.executable == "/usr/bin/python3"` on every new module.

Tests

- `tests/test_ear_feature_representation_audit.py` — 7/7 PASS (representation files present, harness anchor SHAs, feature-cache valset invariance, C3' byte-determinism $\times 2$, verdict-tuple shape, no PRNG, no sidecar_nonfactor imports).
- `tests/test_integration_cross_branch.py` §36 — 12 new checks, all PASS; suite 0 failures overall.
- `promise_check` . — 0 ERRORS, 224 WARNs (this-cycle orphans will adopt on merge via cycle-22 auto-namespacing; pre-existing missing-artifact WARNs unchanged).

Auditor MINOR observations (logged, not investigated)

- Report §1.2 phrasing “`n_files = 84` covering the 55-clip valset + orphans that predate cycle 6” is slightly imprecise given the disclosure that clone-0’s 6 concurrent writes added files during the run. The `feature_cache_pre_post_shas.json` file is authoritative; the narrative sentence could be tightened in a follow-up, but the data is correct.
- Front-matter cycle-6 baseline row uses ($\tau = +0.059$, $\text{MAE} = 0.891$) — a mix of the cycle-22-observed τ with the cycle-6-anchor MAE. Consistent with the brief’s phrasing but a reader could conflate the two contexts; `frontier_summary.json` labels it clearly (variant: `cycle6_baseline`, note: cycle-6 recipe (PC1+noise)). Documentation nit, not a correctness issue.

Discussion

Three things about this branch are worth naming.

First, the closure of Path A is now comprehensive across three orthogonal design axes. Cycle 22 tested the cycle-6 chassis; cycle 23 tested three head-regularization axes (over-fitting, over-parameterization, feature-rank); cycle 25 tested two feature-representation axes (extreme low D at HEUR-only 4, extreme high D at PANNs-only 2048). Six design points, all under the same frozen SHA-anchored / byte-determinism $\times 2$ instrument, all failing the same relaxed C2' bar. The pattern is not a chassis choice, not a regularization choice, not a feature-dimension choice — it is that $N = 55$ synthetic labels do not carry recipe-invariant ordinal information for any reasonable head over any reasonable slice of the frozen cache. This is the strongest possible negative-finding structure without real labels, and it is positive in the sense the campaign should care about: it forecloses further Path A cycles as diminishing information rather than leaving that door open indefinitely.

Second, the HEUR-only C1' PASS is worth preserving as a canonical example of what C2' is designed to reject. The 4-D representation can fit any synthetic ordering the recipes generate because the head has only 4 features to weight against a 55-sample target and the fit picks a different direction per recipe. Mean τ near zero with a symmetric bimodal span $[-0.96, +0.95]$ is the exact fingerprint of “per-recipe overfit, cross-recipe orthogonal” — a MAE improvement over the cycle-6 anchor achieved by fitting orderings that do not compose. C2' was pre-registered specifically to catch this: it asks not “can the head fit some labels well?” but “do the orderings the head learns generalize across recipe perturbations?” The auditor’s report does not spin this as a partial-positive, and future readers should not either — the mechanism is well-understood and the C1'-PASS-C2'-FAIL combination is not evidence that HEUR-only is the right real-label recipe. If anything, the bimodality is a diagnostic hint that

HEUR-only should not be the starting recipe under real labels because the head has too little capacity to hold a consistent ordering.

Third, the two consecutive VALIDATED audits $\times$ two consecutive INVALIDATED verdicts across the four design axes (chassis + three head-regularization + two feature-representation = 6 design points) also validate the stability-audit instrument itself for the third consecutive time. Cycles 22, 23, and 25 each locked a rubric before running, applied it mechanically, and honored the falsifiability contract by publishing FAIL outcomes as first-class findings rather than tuning to force a PASS. The instrument is doing what it should: making the difference between “feature-structured signal on 55 clips” and “recipe-lucky noise fit” legible and pre-registered. This matters directly for the post-egress path — when rated audio unblocks and M-EAR-1/training-loop fires on real labels, the same instrument at the same anchor SHAs is the credibility test, and the real-label success bar must not inherit either cycle-6’s or this cycle’s synthetic thresholds. The frontier plot + report from this branch are the pre-registered “before” evidence against which any real-label success will be judged.

The uncalibrated CORN head under `synthetic_labels_only` remains the campaign’s biggest open credibility gap for the M-GEN-1 scoring pass, and Path B (defer to post-egress real labels) is now the pre-registered outcome. When rated audio arrives via M-INGEST-1/egress-ready-automation, the campaign should start from the cycle-6 chassis with the **original 2052-D features** and not bake cycle-23 or cycle-25 negative findings into the starting recipe — those are chassis-stability findings under synthetic labels at $N = 55$, not statements about the real-label recipe. Whether a HEUR-first or PANNs-first or full-2052-D configuration is right for real labels is a real-label empirical question the synthetic-label instrument cannot answer.

Open Questions

Branch scope is genuinely exhausted. The following are legitimately future-cycle work:

- **Cycle-26 Path B commit** — emit a plan-of-record event superseding any implicit assumption that Path A refinement remained open; commit to real-label ear calibration behind the egress-ready trigger.
- **Cycle-26 anti-patterns to lock:** no 5th regularized head; no further feature slicing; no re-runs of cycle-22 harness with same features + head; no synthetic-label re-audit variants. The two-VALIDATED-audits $\times$ two-INVALIDATED-verdicts $\times$ orthogonal-design-axes structure is the strongest possible negative-finding structure without real labels; additional Path A cycles produce diminishing information.
- **Optional VGGish (R3) closure** — cheap sanity probe if egress remains blocked and cycle 26 has budget. Would either strengthen the Path B commit or unexpectedly reveal a mid-D representation that passes. Low expected information; only if truly cheap. Requires running the VGGish extractor over 55 clips (worker correctly refused to do this in-scope this branch).
- **Post-egress next step** — when `data/ear/rated_ready.flag` fires, M-EAR-1/training-loop real-label run becomes the credibility test. Start from the cycle-6 chassis with the original 2052-D features; do not inherit synthetic-label negative-findings into the real-label recipe.
- **Cosmetic documentation nits** — §1.2 phrasing on `n_files = 84`, front-matter cycle-6 baseline row’s τ /MAE context. Neither is a correctness issue.

Appendix: Provenance

Cycle range: cycle 1 of fork `dc8cba4b79eb`, clone 1. **Working directory:** `/home/user/long-exposure-runs/music-gen`. **Session references:** `researcher 2e0cf2ac-aed1-4d6d-8ac6-63e8a2940fd2`,

worker a94ba8f7-58d3-430e-8f1c-5457c7c3a55e, auditor ac6b885f-c6d7-40d8-bbfd-7b844e7fa596. **Auditor decision: COMPLETE.** Sub-milestone M-EAR-1/feature-representation-audit closes at **invalidated/high** under pre-registered rule 2 (matches the cycle-8 M-TRANS-1/basic-pitch/octave-suppression invalidated/high precedent for negative findings).

Deliverables on disk.

- Code: scripts/ear/feature_representation/{run_representation_audit.py, ...} — interpreter-guarded, no PRNG (AST-checked), no sidecar_nonfactor imports (AST-checked); harness monkey-patched inside try/finally with harness SHA verified pre/post; feature cache read via a valset-scope filter to isolate the operative invariance property.
- Data: data/ear/feature_representation_audit/{variant_verdicts.json, frontier_summary.json, harness_anchor_manifest.json, feature_cache_pre_post_shas.json, stability_report_v3_heur.json (SHA ec429bdf...5e8c), stability_report_v3_panns.json (SHA f98a498c...d39e), per_representation_recipe_details.json, ...}.
- Figures: docs/figures/ear_feature_representation_tau_{mae_frontier,per_representation}.png — frontier includes cycle-6 baseline + cycle-23 3 head-variants + cycle-25 2 representations + R3 deferral marker.
- Report: docs/ear_feature_representation_audit_report.md.
- Test: tests/test_ear_feature_representation_audit.py (7/7 PASS); cross-branch integration test §36 (12 new checks, all PASS).

Load-bearing runtime evidence.

- Two representations, two overall FAIL.
- HEUR-only mean $\tau = -0.076$ (bimodal span $[-0.958, +0.951]$); PANNs-only mean $\tau = +0.006$. Both $\square C2' = 0.4$.
- HEUR-only C1' PASS (best MAE $0.782 < 0.891$ anchor); scientifically interesting, correctly framed as the underdetermined-regressor signature rather than a Path A rescue.
- Byte-determinism $\times 2$ uniform: heur ec429bdf...5e8c, panns f98a498c...d39e.
- Harness anchor SHAs: 6/6 byte-identical to cycle-22 values pre/post.
- Feature cache (valset scope): 55 clips byte-identical pre/post; concurrent-clone all-files interference disclosed and mitigated.
- promise_check: 0 ERRORS.
- Cross-branch integration test §36: 12 new checks, all PASS; suite 0 failures overall.

Ledger routing. Shadow-ledger events emitted at /home/user/music-gen-instance/fork-dc8cba4b79eb/clone-1/promise_ledger.jsonl (plan-register, in-progress checkpoints, terminal M-EAR-1/feature-representation-audit = **invalidated/high**, archive). Canonical UUID5 event_ids per the harness auto-write namespacing fix from fork cc548ca0c2e5 clone 0; nested confidence: {level, rationale, assessor} shape per the SSoT writer. Orphan-artefact WARNs on the new artefacts will clear at post-merge concat via the mechanical adoption pattern.

Environment stack unchanged since cycle 10. mscore3 3.2.3 headless; Python 3.11.15; numpy 1.26.4; music21 9.1.0; mir_eval 0.8.2; fluidsynth (Debian) with pinned SF2 74594e8f...1cb0; DawDreamer + Surge XT Effects.vst3 at /usr/lib/vst3/; basic-pitch 0.4.0 in workspace/basic_pitch_venv/. Single-thread BLAS pins throughout. Stability-audit harness anchors held at cycle-22 SHAs.

Handoff. Merge report at /home/user/music-gen-instance/fork-dc8cba4b79eb/clone-1/merge_report.md. For the root conductor / cycle-26 researcher: commit Path B (defer all ear calibration to post-egress real labels) as the pre-registered outcome; lock the cycle-26 anti-patterns above (no 5th regularized head; no further feature slicing; no cycle-22 harness re-runs with same features + head; no synthetic-label re-audit variants); consider R3 VGGish deferral closure only if cycle 26 has spare budget; when data/ear/rated_ready.flag fires,

M-EAR-1/training-loop on real labels becomes the credibility test — start from the cycle-6 chassis with the original 2052-D features and do not inherit cycle-23 / cycle-25 negative findings into the real-label recipe. The synthetic-label instrument has now exhausted its diagnostic reach at $N = 55$ across chassis, head-regularization, and feature-representation axes.

invalidated